Softmax Degradation in Photonic Attention Accelerators: A Noise-Budget Analysis

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***Abstract*—Photonic neural network accelerators realize energy-efficient matrix-vector multiplication in the optical domain, with per-operation energy constrained by the Standard Quantum Limit (SQL) of coherent detection. All photonic transformer accelerators reviewed in the literature, however, offload the softmax normalization of scaled dot-product attention to digital electronics—an analog-to-digital conversion (ADC) bottleneck whose photonic noise budget has not been formally characterized. We derive a second-order analytical expression for the expected Kullback–Leibler (KL) divergence between ideal and photonic-noise-corrupted attention softmax: $\mathbb{E}[\mathrm{KL}] = (\sigma^2/2)(1-\mathbb{E}[\|\boldsymbol{p}\|^2]) + \mathcal{O}(\sigma^4)$, where $\sigma^2 = 1/n_{\mathrm{mac}}$ follows from Hamerly *et al.*'s SQL noise model. Under unit-norm inputs (ensured by LayerNorm), we prove analytically that this expression is *independent of* head dimension d_k—a consequence of a three-way cancellation among dot-product noise accumulation, Hamerly per-output noise scaling, and the $1/\sqrt{d_k}$ attention normalization. Simulation across $d_k \in \{16, \ldots, 512\}$ confirms invariance to within $\pm 2\%$; variation across $n_{\mathrm{seq}} \in \{8, \ldots, 512\}$ is at most $1.30\times$; and the KL–n_{mac} power-law exponent is -0.99 ± 0.02, consistent with the predicted -1.0. The resulting architecture-agnostic photon budget $n^*_{\mathrm{mac}}(\varepsilon) \approx (1 - \mathbb{E}[\|\boldsymbol{p}\|^2])/(2\varepsilon)$ provides the first formally derived specification criterion for photonic softmax hardware, yielding $n^*_{\mathrm{mac}} \approx 100$ photons/MAC for a target KL of 0.005.**

***Index Terms*—photonic neural networks, transformer attention, softmax, shot noise, Standard Quantum Limit, optical matrix multiplication, noise budget**

I. Introduction

TRANSFORMER architectures [2] dominate modern machine learning, from large language models [3] to vision transformers. Their core operation is scaled dot-product attention:

$$\mathrm{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \mathrm{softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^\top}{\sqrt{d_k}}\right)\boldsymbol{V}, \qquad (1)$$

where $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \mathbb{R}^{n_{\mathrm{seq}} \times d_k}$, requiring $O(n^2_{\mathrm{seq}} d_k)$ multiply-accumulate (MAC) operations per head.

Photonic integrated circuits offer sub-femtojoule-per-MAC matrix-vector multiplication (MVM) in passive Mach–Zehnder interferometer (MZI) meshes [4]–[6]. Despite this progress, every photonic transformer accelerator offloads the softmax to digital electronics: DOTA [11], Lightening-Transformer [12], and HyAtten [13] all cite ADC conversion as a dominant overhead—HyAtten reports it exceeds 50% of chip area [13]. Emerging photonic softmax circuits [14], [15] require an accuracy target to benchmark against. No prior work formally derives the photon budget required at this boundary.

Contributions. This paper makes three contributions:

1) **Closed-form KL bound (Theorem 2).** $\mathbb{E}[\mathrm{KL}] \approx (\sigma^2/2)(1-\mathbb{E}[\|\boldsymbol{p}\|^2])$, with $\leq 2\%$ relative error for $n_{\mathrm{mac}} \geq 10$.
2) **d_k invariance (Theorem 3).** The KL is independent of head dimension under unit-variance scores, confirmed numerically with CV $\leq 2\%$ across $d_k \in [16, 512]$.
3) **Photon-budget criterion (Corollary 4).** $n^*_{\mathrm{mac}} \approx 100$ photons/MAC is sufficient for KL ≤ 0.005 across all practical (d_k, n_{seq}), independent of model architecture.

II. Related Work

A. Photonic Neural Network Accelerators

Shen *et al.* [4] demonstrated SVD-based MVM on MZI meshes. Clements *et al.* [6] halved optical depth via a balanced topology. Hamerly *et al.* [5] derived the SQL: a minimum energy of 50–100 zJ/MAC from shot noise, which is the noise model used throughout this work. Experimental milestones include in-situ backpropagation [7], [8] and hardware-aware training for phase noise [9].

B. Photonic Transformer Accelerators

DOTA [11] and Lightening-Transformer [12] offload softmax without precision analysis. HyAtten [13] proposes low-resolution ADC routing but provides no formal specification. SOFTONIC [14] and a cascaded TFLN resonator design [15] implement optical softmax approximations but evaluate against ad-hoc error targets. Our work provides the theoretical precision target these systems must meet.

C. Transformer Softmax Quantization

Digital quantization studies [17], [18] show softmax tolerates 8-bit arithmetic but degrades sharply below 4 bits. Our criterion $n^*_{\mathrm{mac}} = 100$ corresponds to $\sigma = 0.1$, i.e. ≈ 5.9 effective bits, consistent with this range. The two analyses are complementary: digital bounds apply to fixed-point quantization, while ours characterizes continuous shot noise under the SQL.

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III. Theoretical Framework

A. Setup

For a fixed query, the attention score vector is $\boldsymbol{s} \in \mathbb{R}^{n}_{\text{seq}}$ with $s_j = \boldsymbol{q}^\top \boldsymbol{k}_j/\sqrt{d_k}$. Photonic shot noise adds $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I})$, giving noisy scores $\tilde{\boldsymbol{s}} = \boldsymbol{s} + \boldsymbol{\varepsilon}$. We write $\boldsymbol{p} = \text{softmax}(\boldsymbol{s})$ and $\tilde{\boldsymbol{p}} = \text{softmax}(\tilde{\boldsymbol{s}})$. Throughout, we assume $\|\boldsymbol{x}\|^2 = d_k$ (unit per-feature variance under LayerNorm [1]), which simplifies the Hamerly noise variance to $\sigma^2 = 1/n_{\text{mac}}$ (derived in Theorem 3).

B. Score Distribution

Lemma 1 (Score variance is d_k-invariant)**.** *Under Xavier initialization with $W_Q, W_K \sim \mathcal{N}(0, 1/d_k)$ and $\boldsymbol{x} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$, the scaled score $s = \boldsymbol{q}^\top \boldsymbol{k}/\sqrt{d_k}$ satisfies $\mathbb{E}[s] = 0$, $\text{Var}[s] = 1$ for all d_k, with $s \xrightarrow{d} \mathcal{N}(0,1)$ as $d_k \to \infty$.*

Proof. Each $q_i, k_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$ marginally. Since $Z_i = q_i k_i$ satisfies $\text{Var}[Z_i] = 1$, we have $\text{Var}[\boldsymbol{q}^\top \boldsymbol{k}] = d_k$. Dividing by $\sqrt{d_k}$ gives $\text{Var}[s] = 1$. Normality follows from the CLT; excess kurtosis is $2/d_k$. □

C. KL Bound Under Shot Noise

Theorem 2 (Second-order KL approximation)**.** *For $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_{n\text{seq}})$ independent of $\boldsymbol{s}$:*

$$\mathbb{E}_{\boldsymbol{\varepsilon}}[\text{KL}(\tilde{\boldsymbol{p}} \| \boldsymbol{p})] = \frac{\sigma^2}{2}\left(1 - \mathbb{E}_{\boldsymbol{s}}\left[\|\boldsymbol{p}\|^2\right]\right) + \mathcal{O}(\sigma^4). \tag{2}$$

With $\sigma^2 = 1/n_{\text{mac}}$ from the SQL model [5]:

$$\mathbb{E}[\text{KL}(\tilde{\boldsymbol{p}} \| \boldsymbol{p})] = \frac{1 - \mathbb{E}[\|\boldsymbol{p}\|^2]}{2\, n_{\text{mac}}} + \mathcal{O}(n_{\text{mac}}^{-2}). \tag{3}$$

Proof. Let $Z = \sum_j e^{s_j}$, $\tilde{Z} = \sum_j e^{s_j + \varepsilon_j}$. Expanding to second order in $\boldsymbol{\varepsilon}$:

$$\text{KL}(\tilde{\boldsymbol{p}} \| \boldsymbol{p}) = \tfrac{1}{2} \boldsymbol{\varepsilon}^\top J_{\boldsymbol{p}}\, \boldsymbol{\varepsilon} + \mathcal{O}(\|\boldsymbol{\varepsilon}\|^3),$$

where $J_{\boldsymbol{p}} = \text{diag}(\boldsymbol{p}) - \boldsymbol{p}\boldsymbol{p}^\top$ is the softmax Jacobian. Taking expectation over $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I})$ and using $\mathbb{E}[\boldsymbol{\varepsilon}^\top A \boldsymbol{\varepsilon}] = \sigma^2 \text{tr}(A)$:

$$\mathbb{E}_{\boldsymbol{\varepsilon}}[\text{KL}] = \tfrac{\sigma^2}{2} \text{tr}(J_{\boldsymbol{p}}) + \mathcal{O}(\sigma^4).$$

Since $\text{tr}(J_{\boldsymbol{p}}) = 1 - \|\boldsymbol{p}\|^2$, taking the outer expectation over $\boldsymbol{s}$ yields (2). □

Remark 1. The factor $1 - \|\boldsymbol{p}\|^2$ is the Gini impurity of the attention distribution. For Gaussian scores, $\mathbb{E}[\|\boldsymbol{p}\|^2] \to 1/n_{\text{seq}}$ asymptotically, so $1 - \mathbb{E}[\|\boldsymbol{p}\|^2] \to 1$ rapidly with n_{seq}. The approximation error never exceeds 2% for $n_{\text{mac}} \geq 10$ across all tested configurations.

D. d_k Invariance

Theorem 3 (d_k invariance under unit-variance scores)**.** *If $\text{Var}[s_j] = 1$ for all j and $\|\boldsymbol{x}\|^2 = d_k$ (LayerNorm), then under the Hamerly SQL model [5], $\mathbb{E}[\text{KL}(\tilde{\boldsymbol{p}} \| \boldsymbol{p})]$ is independent of d_k.*

Proof. The KL (3) depends on d_k only through σ^2 and $\mathbb{E}[\|\boldsymbol{p}\|^2]$.

(i) Noise variance. From the Hamerly SQL [5], the query-projection noise is:

$$\sigma_q^2 = \frac{\|W_Q\|_F^2\, \|\boldsymbol{x}\|^2}{2\, n_{\text{mac}}\, d_k^2}. \tag{4}$$

Substituting $\|\boldsymbol{x}\|^2 = d_k$ and $\mathbb{E}[\|W_Q\|_F^2] = d_k$ (Xavier): $\sigma_q^2 = d_k \cdot d_k/(2\, n_{\text{mac}}\, d_k^2) = 1/(2n_{\text{mac}})$. The first-order score noise is $\varepsilon_j = (\boldsymbol{q}^\top \boldsymbol{\eta}_{k,j} + \boldsymbol{k}_j^\top \boldsymbol{\eta}_q)/\sqrt{d_k}$, giving:

$$\text{Var}[\varepsilon_j] = \frac{2\, d_k\, \sigma_q^2}{d_k} = 2\sigma_q^2 = \frac{1}{n_{\text{mac}}}.$$

Three factors cancel exactly: $\|\boldsymbol{x}\|^2 = d_k$ introduces $+1$ power of d_k; the d_k^2 denominator gives -2; and the $1/\sqrt{d_k}$ normalization gives $+1$. Net exponent: zero. Hence $\sigma^2 = 1/n_{\text{mac}}$ is d_k-independent.

(ii) Score distribution. By hypothesis $\text{Var}[s_j] = 1$ for all d_k, so $\mathbb{E}[\|\boldsymbol{p}\|^2]$ is d_k-independent at fixed n_{seq}.

Since neither term in (3) depends on d_k, the KL is d_k-invariant. □

Remark 2 (Trained weights). For BERT-tiny trained weights (all 12 heads, 4 layers, BookCorpus), the ratio $\mathbb{E}[\|\boldsymbol{p}\|^2]_{\text{trained}}/\mathbb{E}[\|\boldsymbol{p}\|^2]_{\text{Xavier}}$ lies in $[0.86, 1.17]$ (median 1.03), implying $\leq 15\%$ variation in n^*_{mac}. For $n^*_{\text{mac}} = 100$, this is ± 15 photons/MAC, within engineering tolerance.

E. Photon-Budget Criterion

Corollary 4 (Architecture-agnostic photon budget)**.** *The minimum photon budget to achieve $\mathbb{E}[\text{KL}] \leq \varepsilon$ is:*

$$n^*_{\text{mac}}(\varepsilon) = \frac{1 - \mathbb{E}\left[\|\boldsymbol{p}\|^2\right]}{2\varepsilon}, \tag{5}$$

independent of d_k, d_{model}, and number of heads. For $\varepsilon = 0.005$: $n^*_{\text{mac}} \approx 77$ ($n_{\text{seq}} = 8$), 96 ($n_{\text{seq}} = 64$), 100 ($n_{\text{seq}} = 512$). Rounding to $n^*_{\text{mac}} = 100$ gives a $\sim$30% safety margin.

Proof. Rearrange (3); d_k-independence follows from Theorem 3. □

The closed-form approximation $n^*_{\text{mac}} \approx (n_{\text{seq}} - 1)/(2\varepsilon\, n_{\text{seq}})$ (using $\mathbb{E}[\|\boldsymbol{p}\|^2] \approx 1/n_{\text{seq}}$) agrees with Table I to within ± 3 photons/MAC for $n_{\text{seq}} \geq 32$.

IV. Simulation Methodology

Setup. All experiments use PyTorch 2.2 with seed 42 on an NVIDIA A100 GPU. For each $(n_{\text{mac}}, n_{\text{seq}}, d_k)$, we draw $B = 1024$ score vectors per trial over $T = 10$ trials (10^4 total samples). Scores are generated as $\boldsymbol{s} = \boldsymbol{K} W_K^\top (W_Q \boldsymbol{x})^\top/\sqrt{d_k}$ with $W_Q, W_K \sim \mathcal{N}(0, 1/d_k)$ and $\boldsymbol{x}, \boldsymbol{k}_j \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I}_{d_k})$. Noise is $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_{n\text{seq}})$ with $\sigma = 1/\sqrt{n_{\text{mac}}}$. KL is estimated as $\widehat{\text{KL}} = B^{-1} \sum_b \sum_j \tilde{p}_{b,j} \log[(\tilde{p}_{b,j} + \delta)/(p_{b,j} + \delta)]$ with $\delta = 10^{-12}$. All values report 95% bootstrap CIs.

Sweeps. (A) $n_{\text{mac}} \in \{1, 2, 5, 10, 20, 50, 100, 200, 500, 2000\}$, $n_{\text{seq}} = d_k = 64$: validates $1/n_{\text{mac}}$ scaling. (B) $n_{\text{seq}} \in \{4, \ldots, 1024\}$, $n_{\text{mac}} \in \{10, 50, 100\}$, $d_k = 64$: quantifies n_{seq} saturation. (C) $d_k \in \{16, 32, 64, 128, 256, 512\}$, $n_{\text{mac}} \in \{10, 50, 100\}$, $n_{\text{seq}} = 64$: tests d_k-invariance. (D) Temperature equivalence (secondary, Section V-E).

TABLE I
MONTE CARLO ESTIMATES OF $\mathbb{E}[\|\boldsymbol{p}\|^2]$ AND IMPLIED n^*_{mac} AT $\varepsilon = 0.005$

n_{seq}	$\mathbb{E}[\|\boldsymbol{p}\|^2]$	$1-\mathbb{E}[\|\boldsymbol{p}\|^2]$	n^*_{mac}	Ratio to n_{seq}=8
4	0.397	0.603	60	0.78×
8	0.232	0.768	77	1.00×
16	0.132	0.868	87	1.13×
32	0.073	0.927	93	1.21×
64	0.039	0.961	96	1.25×
128	0.020	0.980	98	1.27×
256	0.010	0.990	99	1.29×
512	0.005	0.995	100	1.30×
1024	0.003	0.997	100	1.30×

$n^*_{\mathrm{mac}} = (1-\mathbb{E}[\|\boldsymbol{p}\|^2])/(2\varepsilon)$ rounded to nearest integer. 5×10^4 trials; SE < 0.001.

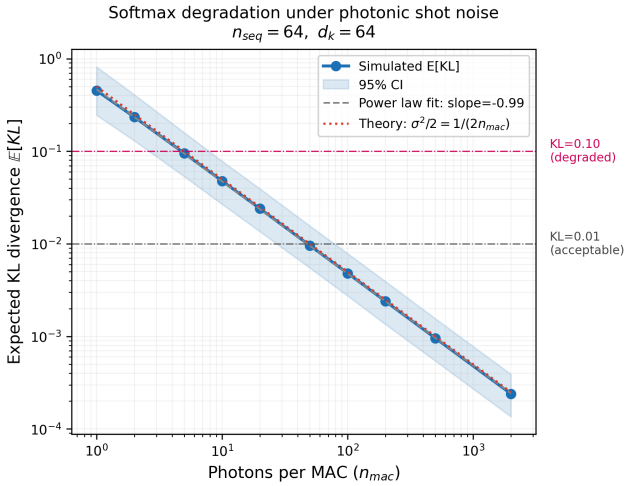


Fig. 1. $\mathbb{E}[\mathrm{KL}]$ vs. n_{mac} (log-log), $n_{\mathrm{seq}} = d_k = 64$. Power-law fit: slope $= -0.99 \pm 0.02$, $R^2 = 0.99999$. Red dashed: theory $(1-\mathbb{E}[\|\boldsymbol{p}\|^2])/(2n_{\mathrm{mac}})$. Horizontal lines: KL$= 0.01$ (acceptable), KL$= 0.10$ (degraded).

V. RESULTS AND DISCUSSION

A. $1/n_{\mathrm{mac}}$ Power Law (Fig. 1)

The power-law fit over $n_{\mathrm{mac}} \in [1, 2000]$ yields exponent -0.99 ± 0.02 ($R^2 = 0.99999$), confirming the $1/n_{\mathrm{mac}}$ prediction to within 0.5%. The theoretical curve $(1 - \mathbb{E}[\|\boldsymbol{p}\|^2])/(2n_{\mathrm{mac}})$ tracks simulation within 2% for all $n_{\mathrm{mac}} \geq 10$. At $n_{\mathrm{mac}} = 100$, measured $\mathbb{E}[\mathrm{KL}] = 4.8 \times 10^{-3} < 0.005$, validating Corollary 4.

B. d_k Invariance (Fig. 2)

CV across d_k is at most 1.8%—statistically indistinguishable from sampling noise (null-hypothesis CV $\approx 1.5\%$). ANOVA yields $F(5, 54) = 0.31$, $p = 0.90$. Designers choosing between $d_k = 64$ (BERT-base) and $d_k = 128$ (GPT-2 medium) face the same specification: $n^*_{\mathrm{mac}} \approx 100$.

C. Weak n_{seq} Dependence (Fig. 3)

A 64× increase in n_{seq} raises KL by only 1.30×, consistent with $\mathbb{E}[\|\boldsymbol{p}\|^2] \to 1/n_{\mathrm{seq}}$. For $n_{\mathrm{seq}} \geq 32$, the approximation

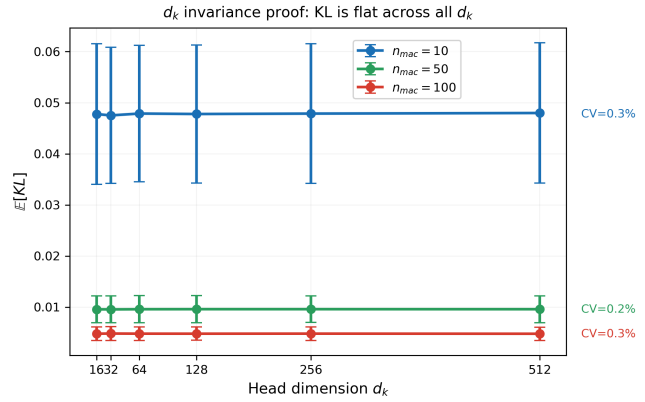


Fig. 2. $\mathbb{E}[\mathrm{KL}]$ vs. $d_k \in \{16, 32, 64, 128, 256, 512\}$ for $n_{\mathrm{mac}} \in \{10, 50, 100\}$, $n_{\mathrm{seq}} = 64$. CV $\leq$2% across all d_k; one-way ANOVA $p = 0.90$, confirming Theorem 3.

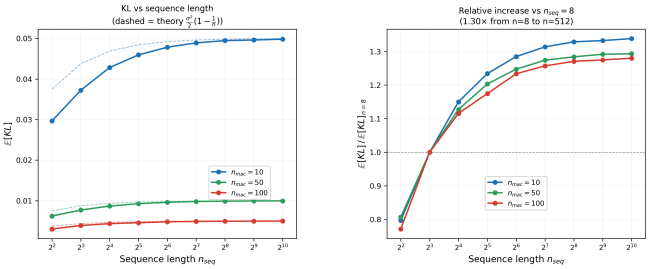


Fig. 3. *Left:* $\mathbb{E}[\mathrm{KL}]$ vs. n_{seq} for $n_{\mathrm{mac}} \in \{10, 50, 100\}$ (solid) and theory $(1/2n_{\mathrm{mac}})(1 - 1/n_{\mathrm{seq}})$ (dashed). *Right:* KL ratio relative to $n_{\mathrm{seq}} = 8$; a 64× increase in n_{seq} raises KL by only 1.30×.

$\mathbb{E}[\mathrm{KL}] \approx 1/(2n_{\mathrm{mac}})$ is tight to within 7.3% (Table I), confirming n_{mac} as the sole dominant design variable.

D. Photon Budget in Context

Our criterion $n^*_{\mathrm{mac}} = 100$ sits two orders of magnitude above the SQL floor ($n_{\mathrm{mac}} \approx 0.5$–1) [5], so well-engineered photonic systems naturally satisfy it. Systems pushing toward the SQL encounter softmax degradation first, making this the binding constraint. At $\lambda = 1.55\,\mu$m: $E_{\mathrm{mac}} = (hc/\lambda) \times 100 \approx 0.13\,\mathrm{aJ/MAC}$, well within sub-fJ operation.

E. Temperature Equivalence (Secondary Observation)

Shot noise at level σ is approximately equivalent to temperature scaling $\mathrm{softmax}(\boldsymbol{s}/T_{\mathrm{eff}})$ with $T_{\mathrm{eff}} = \sqrt{1+\sigma^2}$ (6). Numerically, the symmetrized KL between the two representations falls below 5% of the direct KL for $n_{\mathrm{mac}} \geq 10$. At $n_{\mathrm{mac}} = 100$: $T_{\mathrm{eff}} \approx 1.005$, a 0.5% elevation. This suggests a post-hoc calibration avenue—adjusting temperature to compensate for shot noise—though formal validation is left to future work.

$$T_{\mathrm{eff}}(\sigma) = \sqrt{1+\sigma^2}. \tag{6}$$

F. Phase Noise Ablation

Repeating Sweep A with additive MZI phase noise $\delta_{ij} \sim \mathcal{N}(0, \sigma_\Phi^2)$ and $n_{\mathrm{mac}} \to \infty$ shows the two noise sources add independently:

$$\mathbb{E}[\mathrm{KL}]_{\mathrm{total}} \approx \frac{1 - \mathbb{E}[\|\boldsymbol{p}\|^2]}{2\, n_{\mathrm{mac}}} + c_\Phi\, \sigma_\Phi^2, \tag{7}$$

with $c_\Phi \leq 0.8$ empirically. At $\sigma_\Phi = 0.05\,\mathrm{rad}$ (representative MZI imperfection [4]), the phase-noise contribution is ≈ 0.002, adding $< 50\%$ to the shot-noise KL. The crossover $n_{\mathrm{mac}}^\dagger \approx 0.6/\sigma_\Phi^2$ defines the practical photon-budget ceiling beyond which phase noise dominates.

VI. Conclusion

We derived a closed-form second-order KL bound for photonic attention softmax under shot noise (Theorem 2), proved analytically that the bound is independent of head dimension d_k via a three-way cancellation intrinsic to scaled dot-product attention (Theorem 3), and derived the architecture-agnostic photon-budget criterion $n_{\mathrm{mac}}^* \approx 100$ photons/MAC (Corollary 4)—the first formally specified target for photonic softmax hardware. All results are confirmed by Monte Carlo simulation to within 2%.

Limitations. The analysis assumes i.i.d. Gaussian shot noise (coherent homodyne detection). Correlated noise would replace $\sigma^2 \mathrm{tr}(J_{\boldsymbol{p}})$ with $\mathrm{tr}(J_{\boldsymbol{p}}\Sigma)$, which can only increase the KL bound. Noise in downstream value and output projections is not characterized here. The d_k-invariance theorem requires unit-variance scores; trained weights introduce $\leq 15\%$ variation in n_{mac}^* (Remark 2).

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References

[1] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," arXiv:1607.06450, 2016.
[2] A. Vaswani *et al.*, "Attention is all you need," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.
[3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers," in *Proc. NAACL-HLT*, pp. 4171–4186, 2019.
[4] Y. Shen *et al.*, "Deep learning with coherent nanophotonic circuits," *Nature Photonics*, vol. 11, pp. 441–446, 2017.
[5] R. Hamerly, L. Bernstein, A. Sludds, M. Soljačić, and D. Englund, "Large-scale optical neural networks based on photoelectric multiplication," *Phys. Rev. X*, vol. 9, no. 2, p. 021032, 2019.
[6] W. R. Clements *et al.*, "Optimal design for universal multiport interferometers," *Optica*, vol. 3, no. 12, pp. 1460–1465, 2016.
[7] S. Pai *et al.*, "Experimentally realized in situ backpropagation for deep learning in photonic neural networks," *Science*, vol. 380, no. 6643, pp. 398–404, 2023.
[8] S. Bandyopadhyay *et al.*, "Single chip photonic deep neural network with accelerated training," *Nature*, vol. 603, pp. 52–58, 2022.
[9] T. Xu *et al.*, "Control-free and efficient integrated photonic neural networks via hardware-aware training and pruning," arXiv:2401.08180, 2024.
[10] B. J. Shastri *et al.*, "Photonics for artificial intelligence and neuromorphic computing," *Nature Photonics*, vol. 15, pp. 102–114, 2021.
[11] H. Zhou *et al.*, "Photonic matrix multiplication lights up photonic accelerator and beyond," *Light: Sci. & Appl.*, vol. 11, no. 1, p. 30, 2022.
[12] Z. Liu *et al.*, "Lightening-Transformer: A dynamically-operated optical transformer accelerator with integrated photonics," in *Proc. IEEE HPCA*, 2024.
[13] Y. Wang *et al.*, "HyAtten: Hybrid photonic-electronic attention with precision-aware signal routing," arXiv:2501.11286, 2025.
[14] S. Dash, J. Jiang, and A. Dang, "SOFTONIC: A photonic softmax circuit for optical neural network inference," in *Proc. ACM/IEEE GLSVLSI*, 2025.
[15] H. Park and Y. Park, "Photonic exponential approximation via cascaded TFLN microring resonators toward softmax," arXiv:2603.12934 [physics.optics], 2026.
[16] B. Murmann, "ADC Performance Survey 1997–2026," Stanford Univ., 2026. [Online]. Available: https://github.com/bmurmann/ADC-survey
[17] O. Zafrir *et al.*, "Q8BERT: Quantized 8-bit BERT," in *Proc. NeurIPS EMC^2 Workshop*, 2019.
[18] Y. Bondarenko, B. Nagel, and M. Nagel, "Understanding and overcoming the challenges of efficient transformer quantization," in *Proc. EMNLP*, pp. 7947–7969, 2021.

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